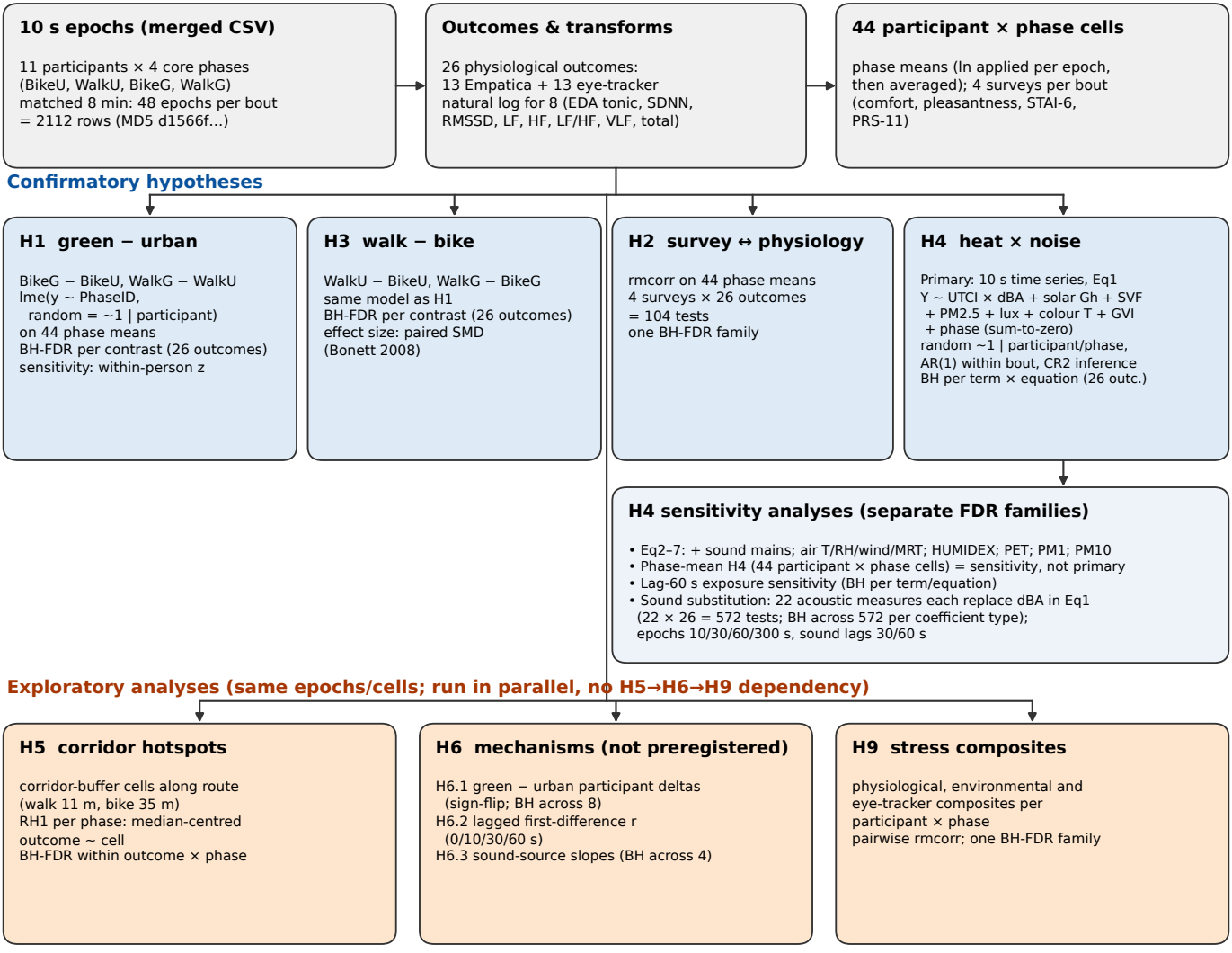
10 s epochs (merged CSV)
11 participants × 4 core phases
(BikeU, WalkU, BikeG, WalkG)
matched 8 min: 48 epochs per bout
= 2112 rows (MD5 d1566f...)
Outcomes & transforms
26 physiological outcomes:
13 Empatica + 13 eye-tracker
natural log for 8 (EDA tonic, SDNN,
RMSSD, LF, HF, LF/HF, VLF, total)
44 participant × phase cells
phase means (ln applied per epoch,
then averaged); 4 surveys per bout
(comfort, pleasantness, STAI-6,
PRS-11)
Confirmatory hypotheses
H1 green − urban
BikeG − BikeU, WalkG − WalkU
lme(y ~ PhaseID,
random = ~1 | participant)
on 44 phase means
BH-FDR per contrast (26 outcomes)
sensitivity: within-person z
H3 walk − bike
WalkU − BikeU, WalkG − BikeG
same model as H1
BH-FDR per contrast (26 outcomes)
effect size: paired SMD
(Bonett 2008)
H2 survey ↔ physiology
rmcorr on 44 phase means
4 surveys × 26 outcomes
= 104 tests
one BH-FDR family
H4 heat × noise
Primary: 10 s time series, Eq1
Y ~ UTCI × dBA + solar Gh + SVF
+ PM2.5 + lux + colour T + GVI
+ phase (sum-to-zero)
random ~1 | participant/phase,
AR(1) within bout, CR2 inference
BH per term × equation (26 outc.)
H4 sensitivity analyses (separate FDR families)
• Eq2–7: + sound mains; air T/RH/wind/MRT; HUMIDEX; PET; PM1; PM10
• Phase-mean H4 (44 participant × phase cells) = sensitivity, not primary
• Lag-60 s exposure sensitivity (BH per term/equation)
• Sound substitution: 22 acoustic measures each replace dBA in Eq1
(22 × 26 = 572 tests; BH across 572 per coefficient type);
epochs 10/30/60/300 s, sound lags 30/60 s
Exploratory analyses (same epochs/cells; run in parallel, no H5→H6→H9 dependency)
H5 corridor hotspots
corridor-buffer cells along route
(walk 11 m, bike 35 m)
RH1 per phase: median-centred
outcome ~ cell
BH-FDR within outcome × phase
H6 mechanisms (not preregistered)
H6.1 green − urban participant deltas
(sign-flip; BH across 8)
H6.2 lagged first-difference r
(0/10/30/60 s)
H6.3 sound-source slopes (BH across 4)
H9 stress composites
physiological, environmental and
eye-tracker composites per
participant × phase
pairwise rmcorr; one BH-FDR family